

Appendix F: Direct-Detection Data Source Notes

Summary file for repository documentation

This document summarizes how the direct-detection inputs used in the collider-versus-direct-detection comparison were prepared. The final comparison uses representative spin-independent exclusion curves from LZ, XENONnT, and PandaX-4T.

Experiment	Input type	Repository file	Treatment in workflow	Reproducibility note
LZ	Machine-readable numerical points	datasets/lz_limit.csv	Imported directly for the sigma_SI overlay plot	Preferred source type because no graphical extraction was needed
XENONnT	Machine-readable numerical points from HEPData	datasets/xenonnnt_limit.csv	Imported directly for the sigma_SI overlay plot	Structured data improved traceability and reproducibility
PandaX-4T	Digitized from published exclusion figure	datasets/pandax4t_digitized.csv	Used to reconstruct the PandaX-4T curve for the overlay plot	Approximate reconstructed curve; limitations documented openly

Digitization summary for PandaX-4T:

1. Identify the published PandaX-4T spin-independent exclusion figure.
2. Calibrate the plot axes in the digitization tool using the mass and cross-section scales shown in the figure.
3. Extract representative points along the PandaX-4T exclusion curve across the mass range relevant to the thesis.
4. Review the extracted points, organize them as a CSV file, and overlay them with the collider-implied, LZ, and XENONnT curves.

Limitation note: The PandaX-4T curve in this repository should be understood as a reconstructed approximation of the published exclusion curve, not an official machine-readable numerical release. Small graphical extraction uncertainties may arise from line thickness, axis calibration, image resolution, and point selection.

Relation to repository datasets: This document corresponds to datasets/lz_limit.csv, datasets/xenonnnt_limit.csv, and datasets/pandax4t_digitized.csv.